

SAN JULIAN, Argentina

WMO: 879090

Lat: 49.3083S

Lon: 67.8031W

Elev: 62

StdP: 100.58

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-4.1	-2.7	-9.8	1.6	9.8	-7.9	1.9	6.1	21.4	6.1	20.5	4.4	5.3	270	0.899

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	27.9	15.8	25.8	14.7	23.9	13.8	16.7	26.1	15.6	24.2	14.6	22.4	11.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.7	9.2	18.5	11.7	8.6	17.7	10.8	8.1	17.1	47.2	26.4	44.0	24.4	41.2	22.4	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.7	15.5	-6.9	32.7	2.0	2.3	-8.3	34.4	-9.5	35.7	-10.6	37.0	-12.1	38.7		
(5)			-7.3	18.8	1.9	1.1	-8.7	19.6	-9.8	20.3	-10.9	20.9	-12.3	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	15.6	15.3	13.5	10.3	6.9	3.8	3.5	5.2	7.8	10.2	12.7	14.7		
	DBStd	5.48	3.27	3.97	3.63	3.42	3.46	3.28	3.27	2.96	3.41	3.51	3.49	3.46		
	HDD10.0	834	1	3	8	37	108	187	202	151	83	40	11	3		
	HDD18.3	3111	96	101	156	242	355	435	458	407	317	252	171	122		
	CDD10.0	812	176	152	116	45	12	2	2	3	17	47	93	148		
	CDD18.3	48	12	17	6	1	0	0	0	0	0	1	3	9		
	CDH23.3	543	151	179	60	4	0	0	0	0	1	8	41	100		
	CDH26.7	128	35	58	11	0	0	0	0	0	0	1	6	18		
(14) Wind	WSAvg	7.7	8.4	8.0	7.2	7.4	6.8	6.9	7.0	7.3	7.6	7.9	8.6	8.6		
(15) Precipitation	PrecAvg	240	17	22	24	18	27	27	25	19	17	19	15	22		
	PrecMax	503	44	77	106	66	66	182	104	60	62	66	49	112		
	PrecMin	115	1	1	0	0	0	1	0	1	0	0	0	0		
	PrecStd	76	11	18	27	16	18	30	28	17	15	13	12	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	32.5	28.0	23.5	18.7	14.5	14.7	16.1	20.8	24.3	27.5	29.0		
		MCWB	16.8	17.5	16.5	14.0	11.8	8.5	8.7	9.5	11.3	12.7	14.9	15.8		
	2%	DB	27.0	28.0	24.9	20.7	16.0	12.0	11.6	13.8	17.4	21.1	24.2	26.0		
		MCWB	15.3	16.0	14.3	12.7	10.0	7.4	7.0	7.9	9.5	11.4	13.5	14.3		
	5%	DB	24.6	25.2	22.2	18.5	14.3	10.2	10.1	12.0	15.7	19.0	22.0	23.8		
		MCWB	14.3	15.0	13.2	11.4	9.1	6.1	6.0	6.8	8.7	10.3	12.1	13.2		
	10%	DB	22.5	22.7	20.0	16.4	12.5	8.7	8.6	10.4	13.9	16.9	19.6	21.7		
		MCWB	13.2	14.1	12.1	10.1	7.9	5.4	4.9	5.9	7.9	9.3	10.8	12.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.7	17.0	14.9	12.4	9.4	9.2	10.5	12.0	13.8	15.6	17.2		
		MCDB	27.6	29.0	27.0	21.8	17.4	13.1	13.7	14.8	18.7	22.6	25.8	27.5		
	2%	WB	16.3	16.8	15.2	13.2	10.8	7.8	7.4	8.5	10.2	12.0	14.0	15.2		
		MCDB	25.2	26.3	22.8	19.7	15.1	11.1	11.2	12.8	16.3	19.5	22.9	24.0		
	5%	WB	15.0	15.7	14.0	11.9	9.6	6.7	6.3	7.4	9.2	10.9	12.7	14.0		
		MCDB	23.3	24.2	20.7	17.4	13.3	9.6	9.3	11.3	14.8	17.6	20.6	22.2		
	10%	WB	14.0	14.5	12.9	10.7	8.4	5.7	5.3	6.2	8.1	9.8	11.5	12.9		
		MCDB	21.3	21.6	18.7	15.6	11.7	8.2	7.9	9.6	13.2	16.1	18.6	20.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.7	10.3	9.4	7.7	6.1	6.3	7.7	9.1	10.4	10.7	10.8		
		MCDBR	14.8	15.4	13.8	11.9	9.9	7.7	8.1	9.9	11.7	13.5	14.4	14.7		
	5% WB	MCWBR	7.1	7.0	6.8	6.8	6.3	5.2	5.3	6.0	6.6	7.1	7.1	6.8		
		MCWBR	13.8	14.2	12.0	11.2	9.0	7.0	7.6	9.4	11.0	12.5	13.6	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.326	0.313	0.306	0.288	0.278	0.277	0.293	0.305	0.312	0.316	0.326		
	taud		2.537	2.543	2.571	2.569	2.567	2.527	2.559	2.560	2.540	2.528	2.529	2.523		
	Ebn at Noon		987	954	914	829	751	704	755	832	907	959	991	995		
	Edh at Noon		104	96	83	68	53	47	52	66	84	97	104	107		
(44) All-Sky Solar Radiation	RadAvg		6.86	5.72	4.01	2.47	1.39	0.95	1.13	2.02	3.36	5.14	6.44	7.04		
	RadStd		0.35	0.23	0.25	0.11	0.07	0.07	0.08	0.13	0.21	0.15	0.33	0.45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page